EART11200: Natural Scientist's Toolkit



Some background on Net Primary Productivity
The 'Greens we eat' problem, and consideration of renewable energy

Overview

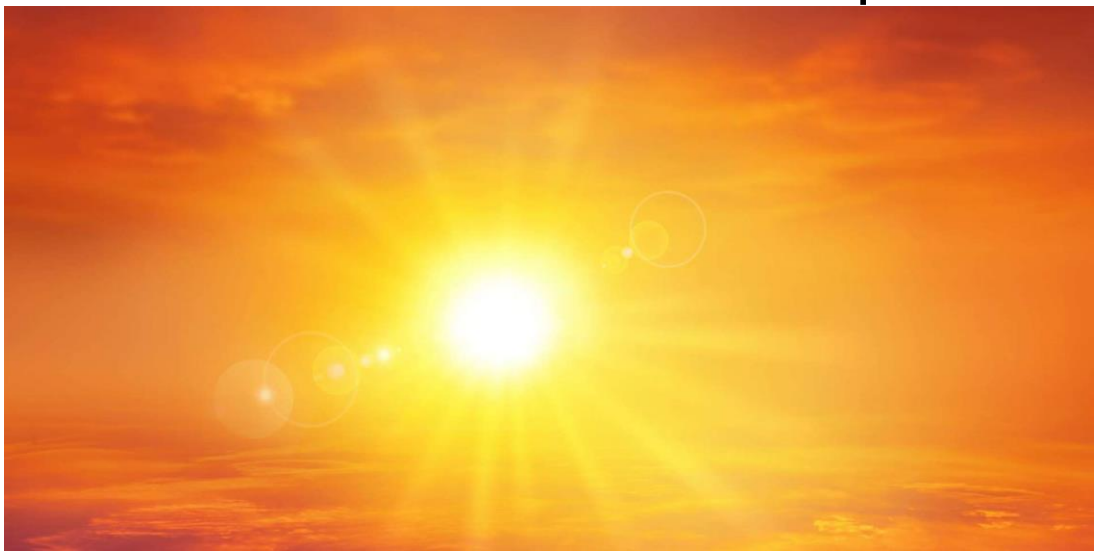
- Primary production is the production of chemical energy in organic compounds by living organisms. The main source of this energy is sunlight but a minute fraction of primary production is driven by lithotrophic organisms using the chemical energy of inorganic molecules

- Regardless of its source, this energy is used to synthesize complex organic molecules from simpler inorganic compounds such as carbon dioxide (CO_2) and water (H_2O). The following two equations are simplified representations of photosynthesis (top) and (one form of) chemosynthesis (bottom):
  - $CO_2 + H_2O + light \rightarrow CH_2O + O_2$
  - $CO_2 + O_2 + 4\ H_2S \rightarrow CH_2O + 4\ S + 3\ H_2O$

- In both cases, the end point is reduced carbohydrate (CH_2O), typically molecules such as ***glucose ($C_6H_{12}O_6$)*** or other sugars. These relatively simple molecules may be then used to further synthesise more complicated molecules, including proteins, complex carbohydrates, lipids, and nucleic acids, or be respired to perform **work (energy)**.

Primary production

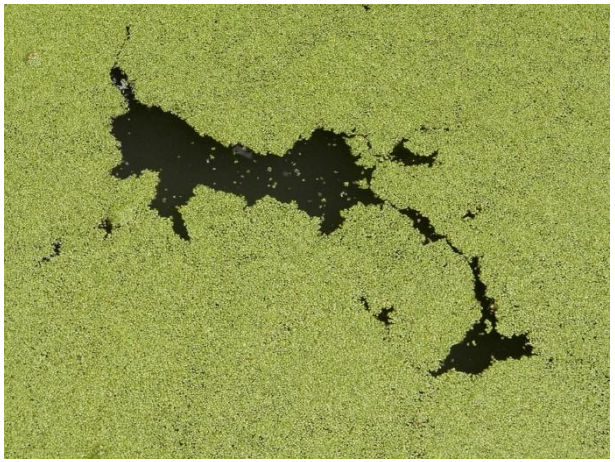
- The production of organic compounds from atmospheric or aquatic carbon dioxide.
- May occur through ***photosynthesis***, using light as a source of energy, or ***chemosynthesis***, using the oxidation or reduction of chemical compounds as energy.



- Almost all life on earth is directly or indirectly reliant on **primary production**. The organisms responsible for primary production are known as primary producers or autotrophs, and form the base of the food chain.

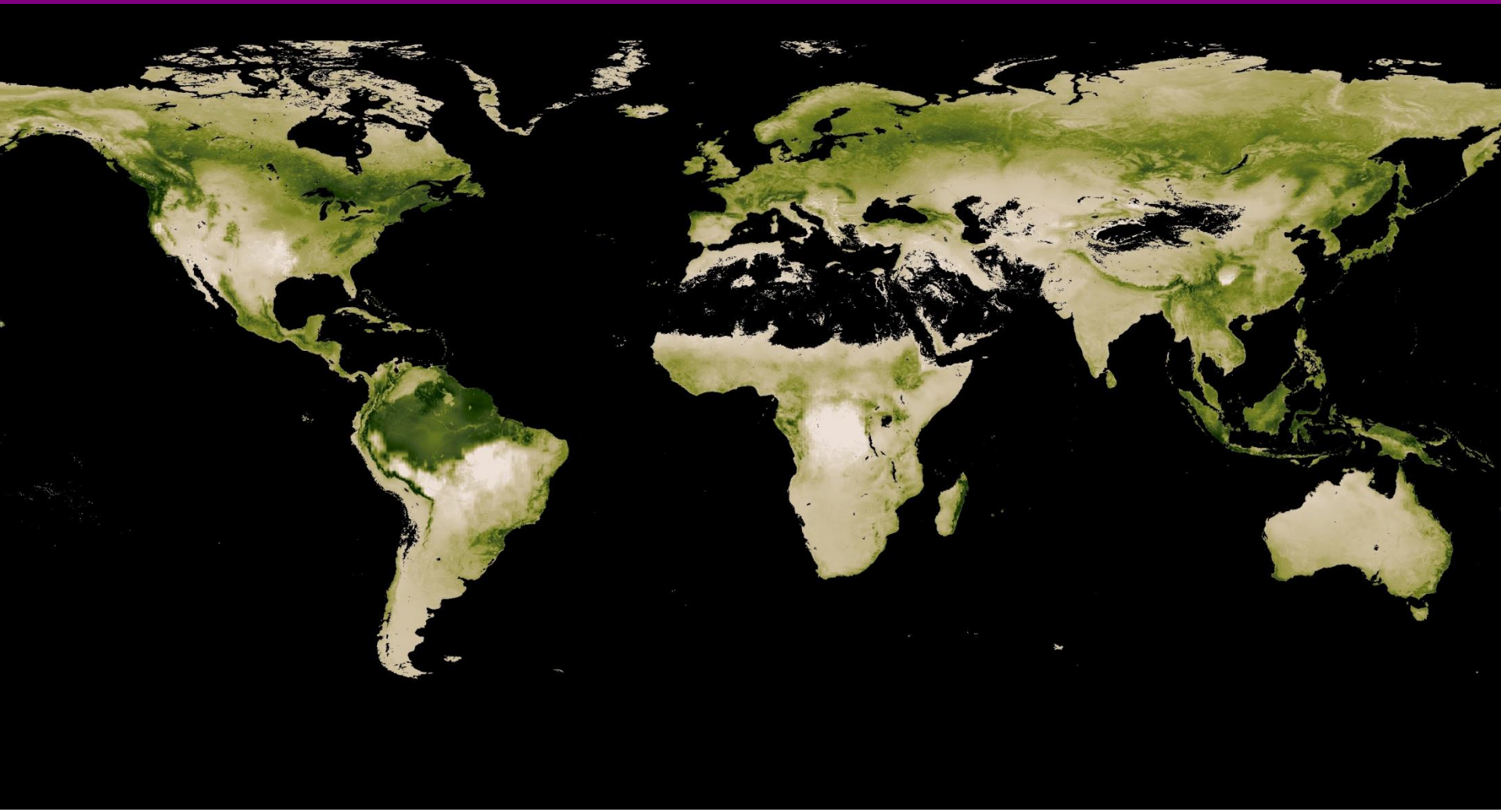
- In terrestrial eco-regions, these are plants, while in aquatic eco-regions it is algae.





- Primary production is either as net or gross, the former accounting for losses to processes such as cellular respiration, the latter not.
- **Net Primary Productivity** can be thought of as the mass of carbon that is converted to biomass.

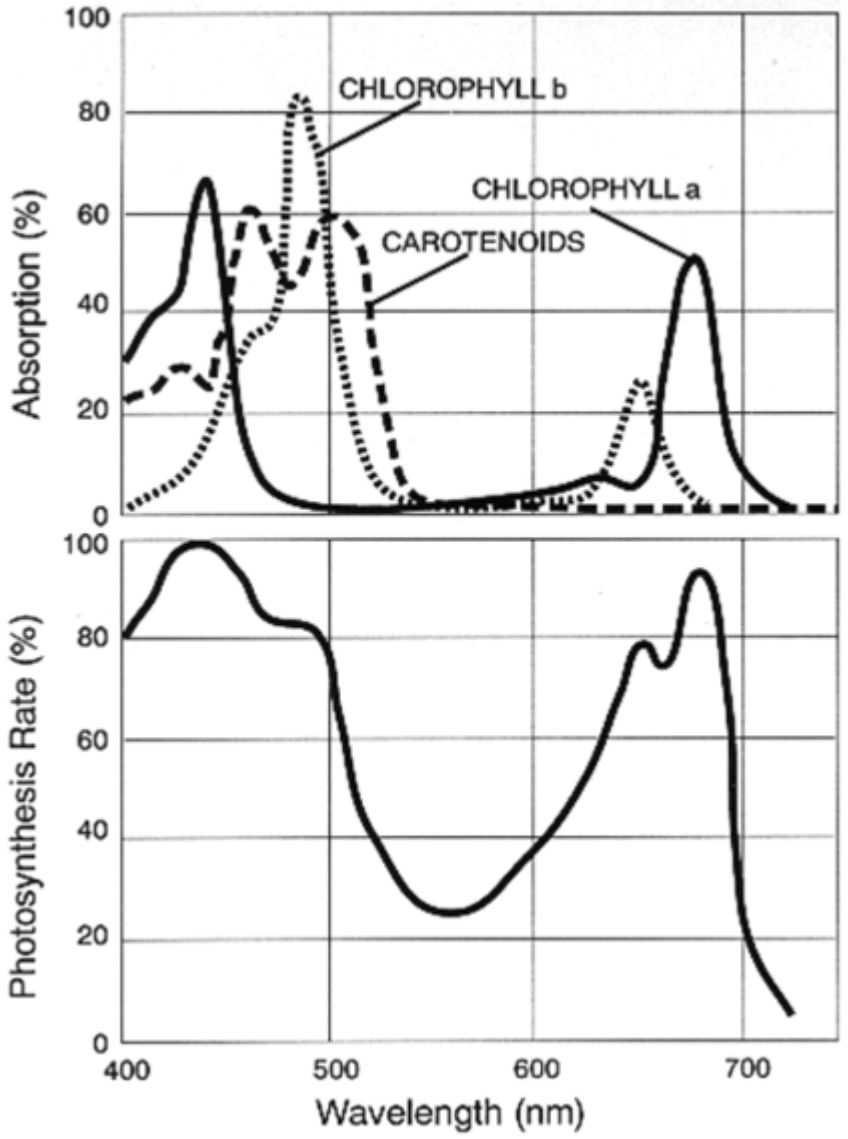
Net primary productivity for August 2011





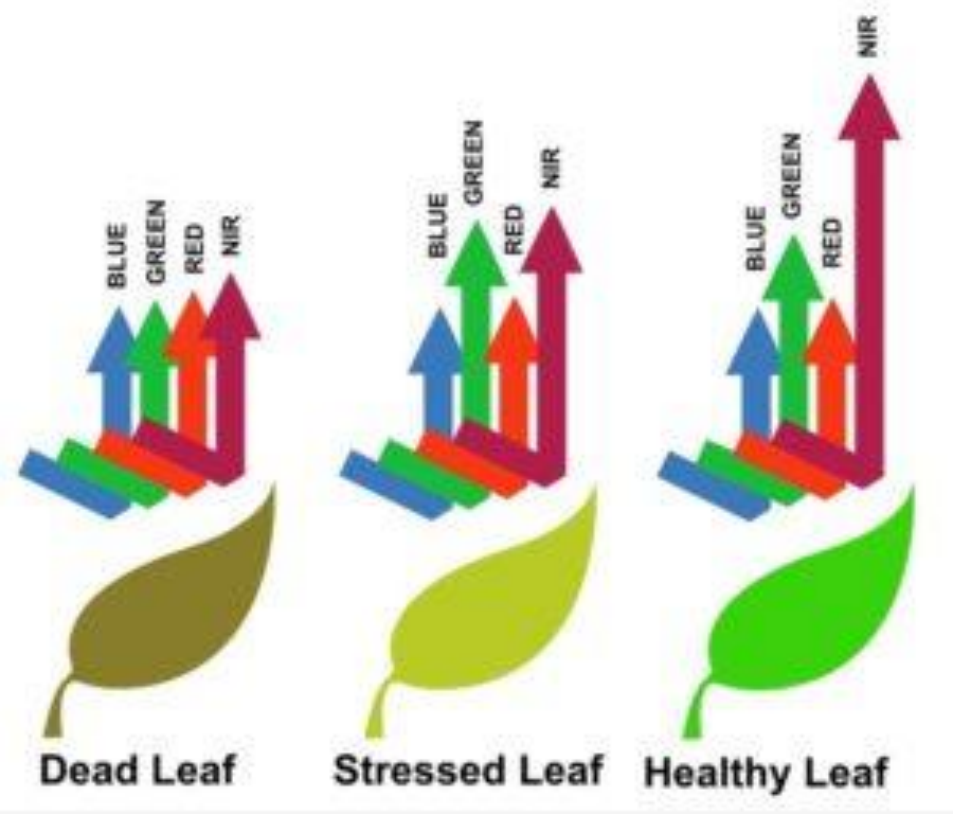
See
https://en.wikipedia.org/wiki/Primary_production
For video from MODIS for 14 years

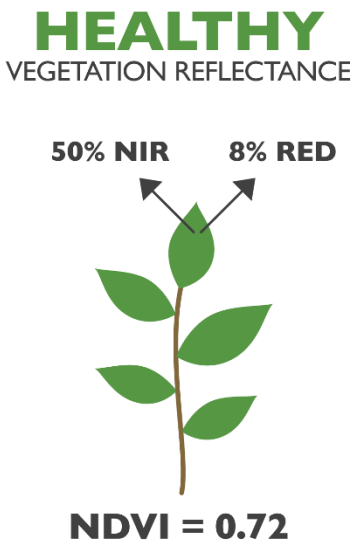
How can we estimate global Primary Production? Well, plants absorb radiation from the sun

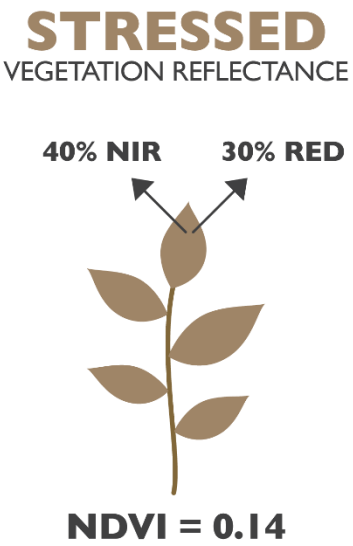


- A surface covered completely by plants will **absorb** visible radiation of different wavelengths.
- This can be related to the photosynthesis rate, which can therefore be used to estimate NPP
- It is more difficult to estimate sub-surface primary production, why?
- How do you think we estimate NPP globally?

How Plants interact with light



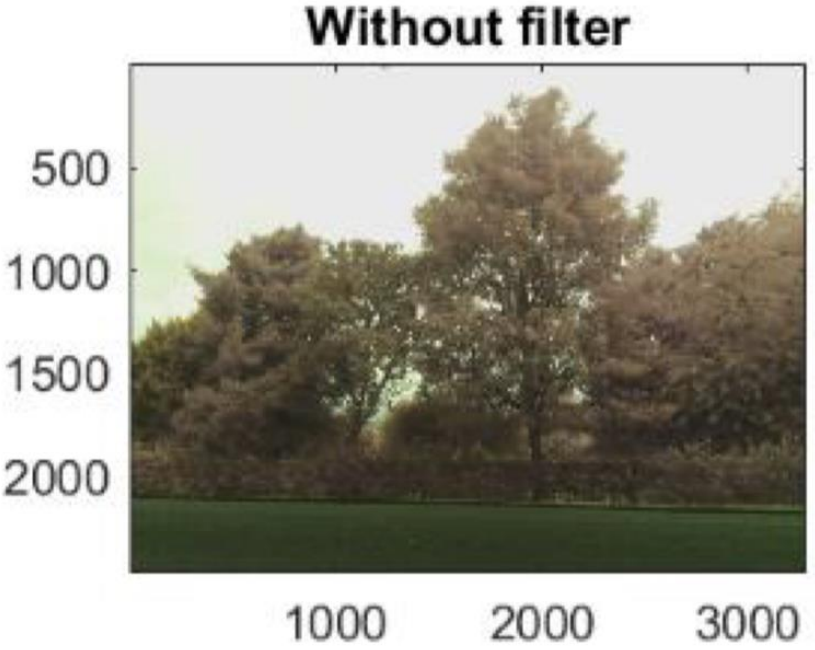


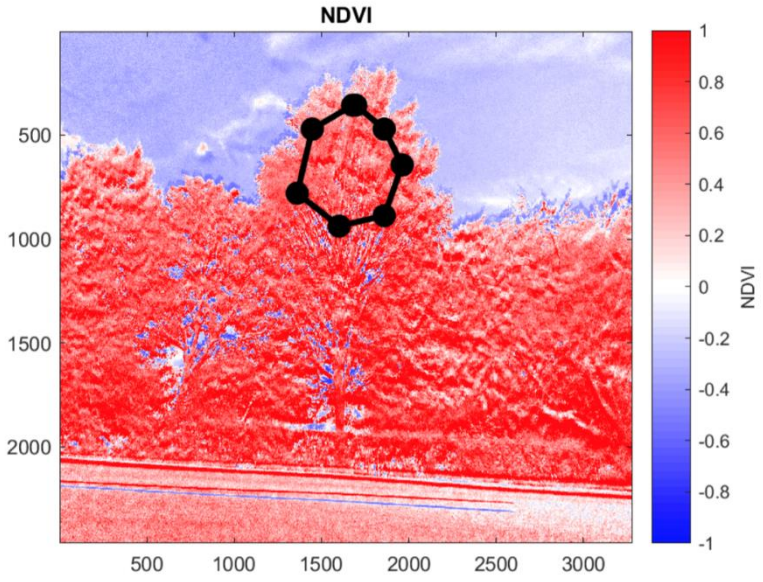


$$NDVI = \frac{NIR - RED}{NIR + RED}$$

Normalized Difference Vegetation Index: NDVI







NDVI=(NIR-red) / (NIR+red)

Plants absorb red light for photo synthesis,

but not near-infrared.

Therefore NDVI is positive

Important point

GPP ≠ NPP

$$NPP = ??\,\mathrm{g\,(C)\,yr^{-1}}$$

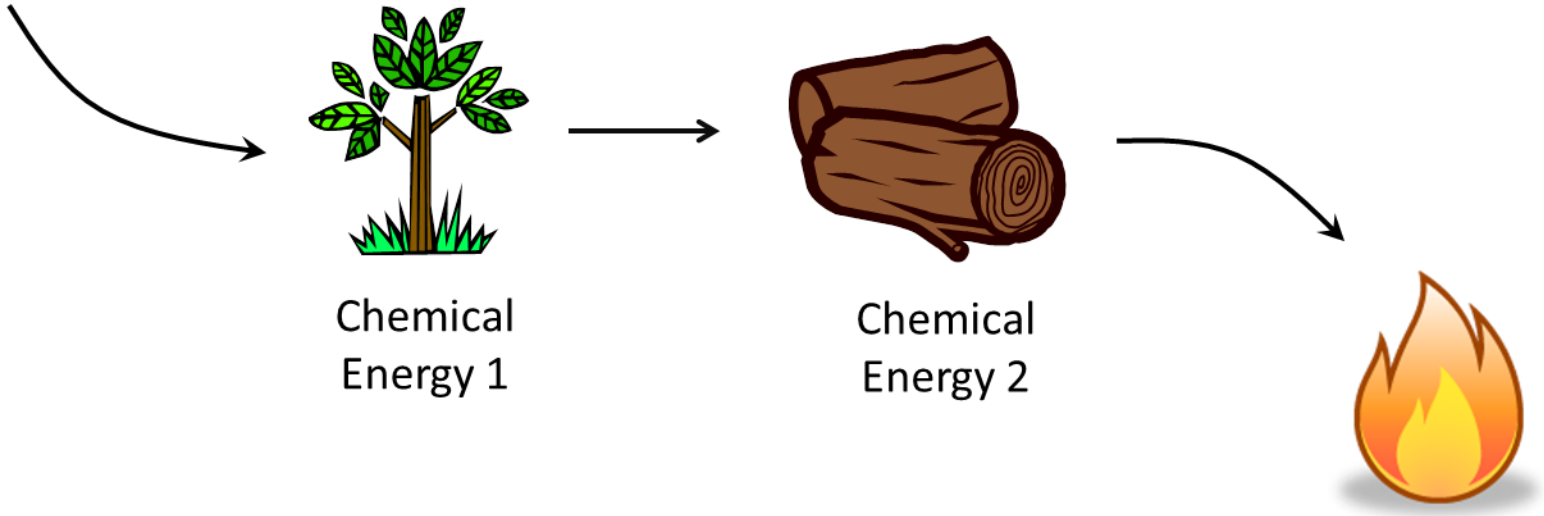
Think of it as the amount of Carbon that goes into plant growth / biomass each year.

It does not equal the mass of the plants growing per year...why?

Carbon/energy stored and then released in combustion

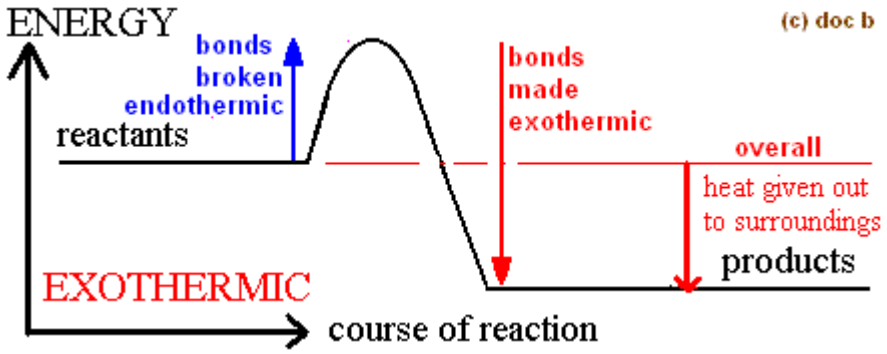
Solar Energy
Visible Light





Heat content of the fuel:

Joules per gram.



Grown Biomass= Energy for power generation

Grown Biomass= Energy for food?





What fraction of Earth's plants need to be burned each year to meet energy demands? Try to guess, and we will calculate it later...



- Can we feed the world?
- Energy – do we need fossil fuels?

https://www.ovoenergy.com/guides/energy-sources/bio-fuels.html

1.3 The greens we eat

Problem 1.3 *The net primary productivity is the amount of elemental carbon that is converted from CO_2 to carbon containing molecules each year and is 14.7PgC (Peta gramms of carbon) per year. If the energy content of dry biomass is 1.6×10^4 J g^{-1} (note J≡joule, which is a unit of energy, see Appendix D) what fraction of the total annual plant growth on Earth was eaten by humans in 2008?*[1]

[1] The worlds population was 6.7×10^9 in 2008

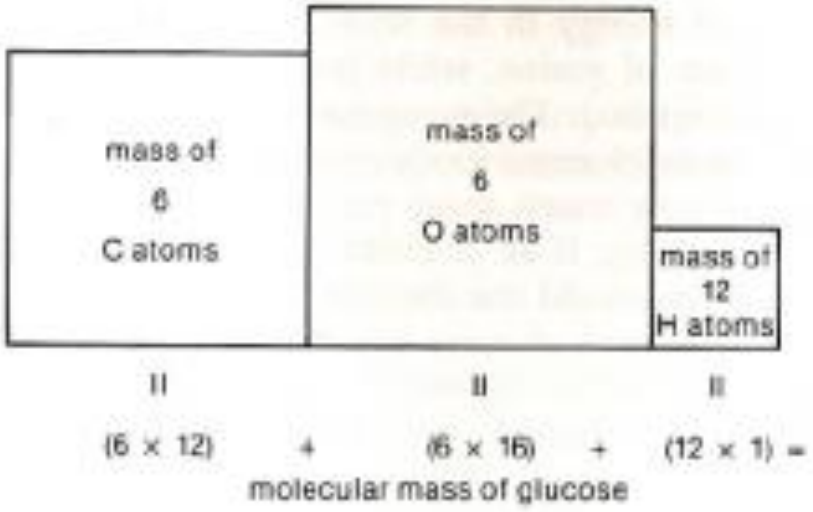


Figure 1.2: The molecular mass of glucose ($C_6H_{12}O_6$) is calculated by adding together the mass of 6 carbon atoms (6 ×12), 6 oxygen atoms (6 ×16), and 12 hydrogen atoms (12 ×1).

You will have to estimate how much energy a typical human eats (e.g. per day). Discuss this with and how you will use it with your group.

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Science & Environment

Beef environment cost 10 times that of other livestock

By Matt McGrath
Environment correspondent, BBC News

21 July 2014 | Science & Environment

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Cattle need 28 times more land than other livestock, according to a new study

A new study suggests that the production of beef is around 10 times more damaging to the environment than any other form of livestock.

https://www.bbc.co.uk/news/science-environment-28409704#:~:text=A%20new%20study%20suggests%20that,any%20other%20form%20of%20livestock.&text=Beef%20cattle%20need%2028%20times,%2C%20poultry%2C%20eggs%20or%20dairy.

In the practical and during week

- First do supplementary material
- Then attempt main problem
- Then related questions.
- In the week do the Blackboard practice assessments. Remember you do need to try these assessments to show engagement, but marks are not counted.